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Taylor & Francis Group

Accounting and Business Research

ISSN: 0001-4788 (Print) 2159-4260 (Online) Journal homepage: http://www.tandfonline.com/loi/rabr20

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Richard D. Morris

To cite this article: Richard D. Morris (1987) Signalling, Agency Theory and Accounting Policy Choice, Accounting and Business Research, 18:69, 47-56, DOI: 10.1080/00014788.1987.9729347

To link to this article: http://dx.doi.org/10.1080/00014788.1987.9729347

Published online: 28 Feb 2012.

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Accounting and Business Research, Vol. 18, No. 69, pp. 47–56, 1987

Signalling, Agency Theory and Accounting Policy Choice*

Richard D. Morris

Abstract—Signalling and agency theories appear in the accounting literature to be competing theories. This article demonstrates that they are actually consistent theories, in that one set of sufficient conditions of signalling theory is at least consistent with one set of sufficient conditions of agency theory. Indeed, a considerable overlap exists between the two theories: rational behaviour is common to both; information asymmetry in signalling theory is implied by positive monitoring costs in agency theory; 'quality' in signalling theory can be defined in terms of agency theory variables; and signalling costs are implicit in some bonding devices of agency theory. Examples are given where both theories' predictions about lobbying, accounting choices, and voluntary auditor selection are added together.

Introduction

In recent years, two seemingly competing theories have appeared in the accounting literature. Agency theory,[1] which describes the incentive problems in a firm caused by the separation of ownership and control of resources (the principal-agent problem) has been used to explain theoretically accounting choices, voluntary disclosure, voluntary appointment of auditors, and corporate lobbying about proposed accounting standards.[2] Signalling theory, originally developed to explain problems of information asymmetry in labour markets,[3] has been applied to questions of corporate dividend policy,[4] capital structure decisions,[5] voluntary disclosure,[6] managerial retention of ownership in new share issues,[7] current value accounting,[8] and voluntary selection of auditors.[9] The literature of agency theory has tended to pay little attention to signalling, and signalling papers ignore agency theory. Yet both theories are used to address similar accounting issues. Consequently, an initial temptation is to consider the theories as competing systems.

This article examines the logical relationship between signalling and agency theories. Far from finding them competing, the conclusion of the article will be that the two theories are consistent: that is, if one theory is 'correct' the other theory may also be 'correct'. This opens up the possibility of joining the two theories to provide fresh insights into the principal-agent problem, and into firms' accounting policy choices.

Agency theory is described briefly in the first section, while an outline of signalling theory appears in the second section. The third section covers the steps involved in examining the relationship between two theories, and these steps are applied to agency and signalling theories in the fourth section. Some accounting applications of the finding that these two theories are consistent appear in the fifth section. Conclusions are given in the last section.

Agency theory

Agency theory is concerned with the principal-agent problem in the separation of ownership and control of a firm (Jensen and Meckling, 1976), between different suppliers of capital (Smith and Warner, 1979), and in the separation of risk bearing, decision making and control functions in firms (Fama and Jensen, 1983). If individuals act self-interestedly, these separations produce conflicts.

Agency theory demonstrates that the precipitators of these conflicts incur agency costs, which they then have an incentive to reduce. The agency costs of equity are, firstly, the decline in a firm's value when shareholders perceive managers as not pursuing the shareholders' interest, especially where managers act inefficiently or do not choose

*The comments of Ray Ball and Peter Birrell are gratefully acknowledged.

[1]Jensen & Meckling (1976).

[2]Reviewed in Holthausen and Leftwich (1983), and in Kelly (1983). Another branch of agency theory reviewed in Demski and Kreps (1982) considers the form of optimal contracts between principal and agent. This literature is not considered here.

[3]Spence (1973, 1974). Other non-accounting applications are in insurance (Rothschild and Stiglitz, 1976) and advertising (Kihlstrom and Riordan, 1984).

[4]Bhattacharya (1979, 1980), Eades (1983), Kalay (1980), Penman (1982a), Miller and Rock (1985), John and Williams (1985).

[5]Ross (1977).

[6]Ross (1979).

[7]Leland and Pyle (1977), Downes and Heinkel (1982).

[8]Forker (1984).

[9]Bar-Yosef and Livnat (1984).

projects as profitable as shareholders would like, and secondly, the costs of monitoring and bonding managers so that they do pursue the shareholders' interest. An optimum trade off exists between these two sets of agency costs. The first agency costs are a manager's opportunity loss, if not reduced by monitoring and bonding, because his self-interested actions precipitate the costs and shareholders build them into the firm's share price.

On the other hand, the agency costs of debt are borne by the equity holders. These costs include (a) the problems of excessive dividend payments, the issue of senior ranking debt, asset substitution and underinvestment (Smith and Warner, 1979), together with bankruptcy and reorganisation costs, and (b) the costs of monitoring and bonding. Rational debtholders incorporate the likelihood of these agency costs in the price they pay for the debt. Again, an optimum trade off exists between monitoring and bonding costs and other agency costs of debt. The agency costs in (a) are opportunity costs to the manager and equity holders if no action is taken to reduce them by monitoring and bonding.

Monitoring and bonding devices include the production of accounting reports, writing restrictive covenants in debt contracts, and management bonus plans geared to reported profits.

There are two version of agency theory. In Jensen and Meckling (1976), an optimum amount of monitoring and bonding is reached beyond which the reduction of each dollar of perquisite consumption costs more than the benefit gained. Consequently, residual agency costs remain. A similar optimum occurs with the agency costs of debt. Extensions of the theory by Fama and Jensen (1983) and Jensen (1983) leave this fundamental result unchanged.

In a second version of agency theory (Fama, 1980), agency costs are driven to zero by market forces within and external to the firm. A form of full ex post settling up occurs which penalises managers who deviate from maximising shareholders' wealth. Thus, they are motivated to act in the interests of shareholders. In essence, Fama's version of agency theory is a long run version of Jensen and Meckling's. For this reason, Jensen and Meckling's version of the theory will be concentrated on here.

Signalling theory

Signalling theory addresses problems of information asymmetry in markets. The theory shows how this asymmetry can be reduced by the party with more information signalling it to others. Although the theory was developed in the labour market, signalling is a general phenomenon applicable in any market with information asymmetry.

In most signalling models, the following steps occur. Initially, sellers in a market are assumed to possess more information about their product than buyers. If buyers have no information about specific products but do have some general perceptions (e.g. that $p\%$ of products offered will be faulty and that bad products should sell at $\$x$ and good products at $\$y > \$x$), then buyers will value all products at the same price which is a weighted average of their general perceptions.

Sellers of above average quality products incur an opportunity loss because their products could sell at a higher price if buyers knew about the superior quality, while sellers of below average products make an opportunity gain. Sellers of high quality products have an incentive to leave the market—the phenomenon of adverse selection (Akerlof, 1970)—unless they can communicate their product's superior quality to buyers and thus increase its price. This communication is done by signalling: the publication of a device (e.g., a product warranty) which acts as a prediction of superior quality. To be effective, the signal must not be easily copied by poor quality sellers. To ensure this, the assumption usually made is that signalling costs are inversely related to quality. Also, the signal must be confirmable with actual product quality observed after purchase.

As better quality sellers signal, buyers consider all the remaining sellers to be of poor quality. Their average price will be reassessed downwards accordingly. The best remaining sellers then try to screen themselves from the others—an iterative process which continues as long as the increase in price obtained exceeds the signalling costs.

The relationship between two theories

The relationship between theories[10] can be assessed in two alternative ways—by examining their underlying axioms and concepts, or by comparing their predictions.

Axiomatic

If two theories deal with the same subject at the same level of reduction,[11] under the axiomatic

[10]A theory is a model or system consisting of (a) non-contradictory, primitive assertions or axioms; (b) definitions of basic concepts; and (c) the conclusions or predictions which are deduced from them. A theory's axioms and definitions are its sufficient conditions (of which there may be several sets)—in their presence the theory's conclusions follow. Necessary conditions, if any, will be part of each set of sufficient conditions. Necessary conditions are axioms which, if negated, mean that the theory's conclusions can never hold.

[11]The level of reduction refers to the level at which a problem is addressed by different disciplines. For example, alcohol addiction is examined as a physiological disorder by physicians, as a behavioural disorder by psychologists, and as a social problem by sociologists. There also can be different levels of reduction within one discipline, such as the classical distinction between micro and macro economics.

approach their relationship is fourfold. Firstly, the theories may be equivalent: the same theory under different guises. Secondly, one theory may imply the other: that is, one is a subset of the other. Thirdly, the two theories may be consistent. This means that if one is true the other is *possibly* true. Fourthly, they may be contradictory or competing explanations of the subject: if one is true, the other is false.

Two steps are needed to investigate which of these four relationships applies to two theories A and B. Firstly, whether or not either theory's axioms are necessary conditions must be identified. To establish the equivalence of theories A and B, both theories' necessary conditions must be identical. Similarly, for theory A to imply theory B, the necessary conditions of B must be the same as those of A, or at least a subset of them. If these stipulations are not met, it is logically possible for theory A to hold in the absence of theory B, a denial of equivalence and implication. On the other hand, consistency of theories A and B only requires that their necessary conditions do not conflict. By contrast, if the two theories are competing, their necessary conditions must conflict in at least one particular.

Secondly, each theory's sets of sufficient conditions must be examined. For equivalence, at least one set of sufficient conditions for A must be identical to at least one set of sufficient conditions of B. For theory A to imply theory B, at least one set of sufficient conditions of B must be entailed in part—but not all—of one set of sufficient conditions of A. Consistency of the two theories only requires that the sets of sufficient conditions of A do not conflict with those of B. However, if A and B are competing theories, at least one set of sufficient conditions for each will be contradictory.

These two steps are illustrated in Figure 1 which depicts, in the upper half, a series of non-contradictory axioms $a, b, \ldots, k$; and in the lower half a set of axioms $\bar{a}, \bar{b}, \ldots, \bar{k}$ which contradict $a, b, \ldots, k$ ($\bar{a}$ contradicts a, $\bar{b}$ contradicts b, and so on). Theories I–V are shown as shapes embracing subsets of these axioms. Axioms $a, b, \ldots, e$ are sufficient conditions for theory I, axioms $e, f, g, \ldots, k$, are sufficient conditions for theory II and so on. Necessary conditions appear as shaded axioms. Theory II implies theories III and IV,[12] while theories I and II are consistent, as are theories I, III and IV. On the other hand, theory V contradicts each of the other theories.

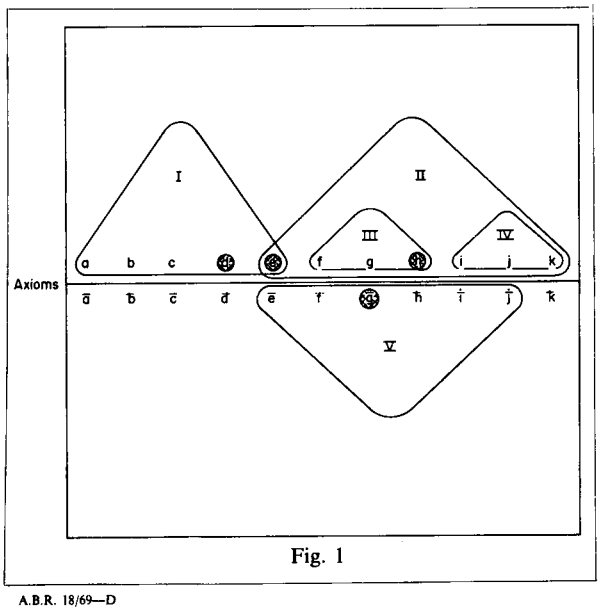


Fig. 1

Prediction

The alternative approach to assessing the relationship between two theories (dealing with the same subject at the same level of reduction) is to compare their predictions or consequences. If these conflict, the theories compete; if they are consistent, so are the theories. (The four way classification of the axiomatic approach is replaced by two categories: competing and consistent.)

As a means of assessing the relationship between signalling and agency theories, prediction comparison is rejected for two reasons. Firstly, comparison of predictions is not reliable if the theories' axioms conflict, because it is possible for the theories' predictions to conflict in one application and to be consistent in another. For example, the Ptolemaic and Newtonian systems of astronomy—competing theories axiomatically—yield the same predictions about planetary motion (and thus in this application are consistent theories) but give conflicting predictions about other astronomical phenomena.

Secondly, in examining the predictions of two theories, the assumption is that the theories' constructs are fully operationalised. At present, this is not so with signalling and agency theories. For example, in agency theory, leverage is used as a proxy for closeness to accounting-based borrowing limits, and bonus plans are usually represented in empirical studies by a dichotomous variable. In signalling theory, quality could be measured using next year's change in reported earnings or share market abnormal performance. However, all of these proxies are imperfect:[13] they measure the

[12]In some investigations of the relationship between two theories, the approach adopted is to demonstrate that the two theories are special cases of a general or meta theory. The general theory implies the special case theories. In Figure 1, this is exemplified by theories II, III and IV. Yetton and Crouch (1983) provide an example of this approach in the leadership literature.

[13]Ball and Foster (1982) outline the weaknesses of proxy measures in agency theory. Abnormal share price performance and net profit changes next year are imperfect proxies of quality because, although they capture quality changes expected now by managers but known to investors later, they also capture the effects of unforeseen events arising in the ensuing year.

underlying theoretical construct with error. A difficulty created is that predictions using imperfect proxies may compete (or be consistent) due to this measurement error, leaving unclear whether the underlying theories compete or not.

The remainder of the paper concentrates on assessing the relationship between agency and signalling theories using the axiomatic approach.

The relationship between agency theory and signalling theory

In the first section an outline of agency theory was given. The following assumptions constitute a set of sufficient conditions from which the features of that theory can be derived.[14]

(1) All market participants are rational wealth maximisers.
(2) All firms operate in two periods.[15] Managers make production/investment decisions in the first period which affect firms' expected values and variances in the second period.
(3) Firms have external equity and debt financing.
(4) There is separation of equity and debt capital suppliers and managerial control in the firm.
(5) Each manager owns a fraction[16] of his firm's outstanding equity.
(6) Each manager is remunerated by salary, perquisite consumption, and returns on his equity in the firm.
(7) Monitoring and bonding are available at a cost proportional to the value of the firm; and they reduce activities dysfunctional to capital suppliers at a reducing rate.

In the second section an outline of signalling theory was given. From it, the following set of sufficient conditions for signalling in a capital market setting may be drawn.

(a) All market participants are rational wealth maximisers.[17]
(b) All firms operate in two periods. Managers make production/investment decisions in the first period which affect the quality of firms in the second period.
(c) The quality of firms competing for equity and debt funds in the capital market varies.
(d) The actual quality of firms is objectively observable, ex post, in period 2.
(e) Ex ante, in period 1, information asymmetry exists between the manager of each firm and capital suppliers. The manager's information about the firm is superior to that of capital suppliers.
(f) Signalling costs are inversely related to quality.

The relationship between each theory's assumptions will now be discussed.

Necessary Conditions

Individual rational wealth maximisation (assumption 1) and separation of resource ownership and control (assumption 4) are necessary conditions of agency theory. Together, they imply conflicts of interest between managers of firms and capital suppliers. Without them, the principal-agent problem would not exist. Similarly, information asymmetry (assumption e) is a necessary condition of signalling theory, because without it the need for signalling does not exist.

Prima facie, these necessary conditions of each theory do not conflict. Positive monitoring costs (assumption 7) and separation of ownership and control (assumption 4) imply information asymmetry between managers, investors and creditors. Thus, information asymmetry is implicitly contained in the sufficient conditions of agency theory given above. However, positive monitoring costs are not a necessary condition of agency theory, because the principal-agent problem can still exist in the (unlikely) case of zero monitoring costs, provided that bonding costs are positive. The latter ensures that residual conflicts remain even if all parties are fully informed.

Therefore, agency and signalling theories do not share common necessary assumptions. Hence, the theories are not equivalent, nor does one imply the other.

Sufficient Conditions

The first four assumptions of both theories, given above, are compatible except that agency theory does not mention the concept of 'quality'. However, because signalling is a general phenomenon applicable in any market with information asymmetry, the meaning of 'quality' can be taken from the market context being considered. Therefore, quality may be defined in terms of a firm's expected value, its risk, and the level of the manager's compensation: three variables directly addressed by assumptions (2) and (6) of agency

[14]For brevity, definitions of basic terms such as monitoring and bonding are omitted.

[15]These may be thought of as the present for period one, and the future for period two.

[16]In Jensen and Meckling (1976) this fraction commences at 100 percent and is reduced as the manager sells part of his equity to outsiders.

[17]Some might object that this assumption is at variance with other applications of signalling theory in capital markets (e.g., Ross, 1977; Bhattacharya, 1980; Kalay, 1980; Talmor, 1981) which assume that managers act in the interests of investors. However, that assumption is not crucial, and is usually made for convenience. The original exposition of signalling theory by Spence (1973, 1974) had market participants acting in their own interests, as did later applications of the theory by Rothschild & Stiglitz (1976), Leland and Pyle (1977), Bhattacharya (1979), Haugen & Senbet (1979), Downs & Heinkel (1982), Heinkel (1982), and Eades (1983).

theory.[18] The manipulation of these variables can produce agency costs. Thus, an above average quality firm will have a higher expected value than firms of comparable risk and levels of management compensation.

Information asymmetry (assumption e) of signalling theory is implied by positive monitoring costs (assumption 7) and the separation of ownership and control of capital (assumption 4) in agency theory, as stated previously.[19] This information asymmetry leads to costs in the form of opportunities foregone by the manager of an above average firm for raising equity or debt capital. Less will be paid for the firm's equity or debt securities than would be the case with no information asymmetry.[20] As losses from foregone opportunities, these costs are identical to agency costs of equity or debt. In short, the costs of information asymmetry are a subset of the construct agency costs. As with other agency costs, these costs of information asymmetry are borne by the manager of an above average firm. Graphical proofs of this appear in the Appendix. The manager then has an incentive to signal his firm's above average quality to reduce this opportunity loss.

However, if the firm is below average, the agency costs of information asymmetry are borne by investors since, in the absence of signalling, the market regards the firm as average quality. This conclusion alters a central result of agency theory: that agency costs are always born by the agent. With information asymmetry, some agency costs of below average firms are borne by investors.[21]

For signalling and agency theories to be consistent, signalling costs (assumption f) must be borne by the agent so that he has an incentive to signal truthfully. Also, these signals should not be inconsistent with monitoring and bonding devices (assumption 7) in agency theory. That these conditions hold can be illustrated by reference to three common bonding devices in agency theory.

(1) Management compensation plans are bonding devices because they align the interests of the manager and the shareholder (Jensen and Meckling, 1976). Many formal management compensation plans specify some threshold profit level below which no bonus is paid to managers (Healy, 1985). This threshold will be negotiated by each manager and his shareholders. At the point of negotiation, the threshold acts as a prediction of attainable future earnings since the manager acting in his own interest will want as much bonus as possible. Better quality managers may negotiate a higher threshold to discriminate themselves from poorer quality managers. If a manager believes his firm to have a higher than average expected value, he will negotiate a compensation package, which, for example, contractually sets the maximum bonus to be paid if the expected value is attained. Also, to reduce the resulting horizon problem in which projects with high short term payoffs but not necessarily high net present values are chosen, above average firms will gear the bonus to some measure which takes account of all expected cash flows, e.g. share price or average earnings. So, bonus schemes can act as a signal.

(2) Contractual debt covenants restricting the amount of debt a firm can raise limit wealth transfers between equityholders and debtholders. Managers have an incentive to offer these protective debt covenants to increase the price at which debt is sold, and indirectly to act as a signal about expected future earnings and expected levels of management compensation. Ross (1977) shows that where managerial compensation contracts are in the form of a contingent contract and where opportunities for wealth transfers to the equityholders are absent (or perfectly controlled), the contractual level of debt issued signals the expected value of the firm. Firms with higher contractual debt equity ratios have above average expected values. Maximum contractual debt equity ratios can also signal limits to a manager's inefficiency (perquisite consumption) because his inefficiency will adversely affect shareholder's equity on the balance sheet (e.g., by reducing reported earnings) and raise the debt equity ratio towards its limit. So, contractual debt equity ratios can act as signals as well as bonding devices.

(3) Dividend constraints are bonding devices because they restrict the manager's ability to transfer wealth to the equityholders. A manager's incentive thus to pay dividends is positively related to his firm's leverage ratio (Kalay, 1978). Since the manager (and the equityholders) bear the resultant agency costs, he will attempt to reduce these by offering protective debt covenants, such as the dividend constraint. Other things being equal, the greater the agency costs of debt, the tighter such a constraint will be. However, a tight constraint increases the chance of forced investment in negative net present value projects, since the dividend constraint may forbid distribution of surplus funds

[18]Quality may also be defined in terms of other variables such as turnover, plant efficiency etc. However, it is sufficient for the purpose of examining if agency and signalling theories are consistent to show that 'quality' can be interpreted in terms of agency theory variables.

[19]Jensen & Meckling (1976) ignore information asymmetry, claiming that it will not affect their theory in large equity markets and where estimates are rational and errors are independent across firms (p. 318). However, because of the incentive for adverse selection to operate where information asymmetry is present, measurement errors will not be independent across firms (Ronen, 1979).

[20]Barnea, Haugen and Senbet (1981) and Weston (1981) also suggest that information asymmetry produces these costs.

[21]This observation was first made by Ronen (1979). However, he did not draw out the difference in who bears agency costs between above and below average firms.

to shareholders. This is an overinvestment problem.

Firms with high growth potential, that is a large number of positive net present value projects available, will find this overinvestment problem less costly than other firms and so will tend to offer tighter dividend constaints to debtholders. In this way, a tight dividend constraint can act as a signal about the expected value of the firm.

Signalling Costs

Each of these three bonding devices has a signalling cost function entailing opportunity costs. The manager incurs no penalty if he does not deviate adversely from the contractual levels set by the three devices. (There are also direct contract negotiation costs contained explicitly in agency theory.) If adverse deviation from contract does occur, the manager is penalised the more he deviates. These penalties include a decline in his 'price' in the market for managers and wealth opportunities foregone for the firm and for the manager due to his lack of diligence.

Fama (1980) explicitly models these penalties in his version of agency theory as the process of full ex post settling up. However, Jensen and Meckling (1976) are silent about these penalties. Nevertheless, their version of agency theory is not inconsistent with the existence of such penalties. Indeed, Jensen and Meckling discuss the role of factors other than direct monitoring and bonding outlays (p. 328) in controlling the behaviour of managers, but do not include them in their model.

Conclusion

Since the sufficient conditions outlined for signalling theory are at least consistent with those of agency theory, it can be concluded that the two theories are consistent. Indeed, a considerable amount of overlap exists between the theories because several of the sufficient conditions for signalling are implicit in agency theory. However, because information asymmetry, a necessary condition of signalling theory, is not a necessary condition of agency theory, signalling theory is not implied by agency theory; nor are they equivalent theories.

Applications

From the previous section, it is clear that some constructs of agency theory—bonus plans, contractual debt limits, and dividend constraints—can also act as signals. An implication for accounting research is that proxy variables used to measure these constructs may capture both their agency theory and signalling aspects. Results of empirical studies using these variables must be carefully interpreted as a result.

Given the consistency of signalling and agency theories, it is conceivably possible to combine them to yield predictions about accounting choices not obtainable from either theory alone.[22] However, the problems of operationalising both theories, mentioned earlier, make this a difficult task. At present, the prediction of accounting choices can at least be improved by adding together the predictions from each theory. (This is possible logically given the consistency of the theories.)[23] Some examples follow.

Corporate Lobbying

Existing studies[24] of corporate lobbying about proposed accounting standards have used agency theory to predict the decision to lobby and lobbying position. Larger firms tend to lobby. There is also some evidence that the decision to lobby is associated with leverage and the use of management bonus schemes. Larger firms oppose accounting standards which increase reported earnings, while firms with accounting based debt or management compensation contracts favour such standards. The opposite applies to proposed standards which reduce reported earnings.

Generally accepted accounting principles set a lower bound on the fineness of disclosed accounting information. They also contain a degree of flexibility which is often criticised in the accounting literature. Rational investors and creditors will expect management to use this flexibility in reporting the firm's performance in the most favourable way. Therefore, a tendency exists here for adverse selection to operate. To counteract this tendency, high quality firms will wish to signal that they are not using this flexibility to mislead shareholders, or that they are not doing so as much as other firms. One way of doing this is for such firms to be seen as users of the 'best' accounting policies advocated by the accounting profession, provided that these policies do not allow low quality firms to masquerade as high quality ones. Therefore, it is in the interest of high quality firms that accounting principles allow them to reveal that they are in fact high quality.

To ensure that accounting standards of this kind are introduced, high quality firms will lobby the accounting profession. So too will poorer quality firms, in whose interest it is that accounting standards do not provide fine information signals. So both types of firm have an incentive to lobby.

[22]Such a 'meta-theory' could also contain new concepts not included in either signalling or agency theories.

[23]For example, Puro (1984) argues that agency theory and the economic theory of regulation are consistent, and takes together the predictions from these theories to explain audit firms' lobbying practices.

[24]These studies have investigated corporate lobbying about general price level accounting (Watts & Zimmerman, 1978), foreign currency translation (Griffin, 1982, 1983; Kelly, 1982) and interest capitalisation (Dhaliwal, 1982).

Lobbying itself is not a signal, mainly because there seem to be no penalties for making a misleading lobbying submission.

For both high and low quality firms, the benefits from lobbying should be higher the greater the information asymmetry between the firm and its investors. This will occur in firms with large numbers of shareholders or geographically widespread shareholdings.

Accounting Policy Choice

In several studies,[25] accounting policy choices were related to size and debt covenants, and to a lesser extent to management compensation schemes. As with lobbying, larger firms favour accounting policies which reduce reported earnings, while firms with debt covenants and management compensation schemes based on reported earnings favour income increasing accounting policies, ceteris paribus. The contribution of signalling theory is the prediction that higher quality firms will choose accounting policies which allow their superior quality to be revealed, while lower quality firms will choose accounting methods which attempt to hide their poor quality. For example, a higher quality firm may voluntarily adopt segment reporting to disclose the superior risk/return profile of its operations, but a low quality firm would not. Similarly a high quality firm may voluntarily disclose an earnings forecast, but a low quality firm would not. As in lobbying, the incentive to signal by accounting policy selection should be highest where information asymmetry is greatest: in firms with large and widespread numbers of shareholders. Nevertheless, the number of accounting policies which act as valid signals may be limited, given the present flexibility of generally accepted accounting principles.

Voluntary Auditor Selection

Chow (1982) demonstrated how leverage, accounting based debt covenants and firm size were related positively to voluntary auditor appointments by US companies when such appointments were unregulated. Signalling theory's prediction is that auditors will be appointed voluntarily by higher quality firms in order to discriminate themselves from other firms in the market. Again, this incentive should be greatest in firms with large and widespread numbers of shareholders. Further aspects of signalling and auditing are discussed in Bar-Yosef and Livnat (1984).

Summary and conclusions

Any two theories at the same level of reduction may be either equivalent, consistent or competing accounts, or one theory may be a subset of the other. Agency theory and signalling theory are not equivalent, nor does one theory imply the other, because they do not share the same necessary conditions. However, the sufficient conditions of signalling theory given in the fourth section are at least consistent with those of agency theory. Rational behaviour is common to both theories; information asymmetry is implied by positive monitoring costs in agency theory; quality can be defined in terms of agency theory variables; and signalling costs are implicit in some bonding devices of agency theory. Therefore, agency theory and signalling theory are consistent. Indeed, a considerable amount of overlap exists between them.

The fifth section offered examples where both theories' predictions about lobbying, accounting choices, and voluntary auditor selection were added together (logically possible with consistent theories). However, further research is needed to combine signalling and agency theories into a more general 'meta-theory' which might yield insights into the principal-agent problem, and choice of accounting methods, not obtainable from either theory alone.

Appendix

The objective of this appendix is to demonstrate that information asymmetry can lead to agency costs which are borne by the agent in above average firms, and by the principal in below average firms. The graphical device employed has been used previously by Myers (1977), Kalay (1978) and Holthausen (1979) to demonstrate the wealth transfers that can occur between debtholders and equityholders in a firm.

Assumptions

1. Firms operate in two periods, t_0 = the present and t_1 = the future. Figures 2 and 3 show t_1 outcomes in t_0 present values.
2. There is no debt financing.
3. Where no information about specific firms is available, they are treated alike by investors, whose general perceptions are determined by exogenous factors such as their general knowledge about the corporate sector, the state of the economy etc.
4. Risk about the value of firms is depicted as varying t_1 outcomes in different states of the world.
5. All states of the world are equally likely to occur.
6. For convenience, states of the world have been arranged in Figures 2 and 3 so that value schedules are linear.
7. Firms are wholly owned by managers who

[25]Holthausen and Leftwich (1983) and Kelly (1983) review these studies.

intend to float shares in them to outside investors.

8. Managers maximise their wealth by maximising their expected compensation and the price at which shares are issued.
9. Each manager's compensation is a fixed proportion of the expected value of his firm in t_1. This compensation has priority of payment over returns to investors.
10. Signalling costs are inversely related to the value of the firm in any state of the world. Signalling costs rank after management compensation in priority for payment.
11. All firms have accepted projects with net present values of at least zero in all states of the world.

These assumptions differ slightly from those in the body of the paper in order to streamline the exposition which follows.

Figure 2 shows $A\bar{V}$, the value of all firms where no signalling or monitoring is permitted. Investors perceive all firms as having the same expected value $E(\bar{V})$ and risk. Managers' compensation is negotiated as $\alpha E(\bar{V})$ shown as MBC. Since all states of the world are assumed equally likely to occur, the value of managers' compensation and the value of equity can be depicted as segments OABCD and $B\bar{V}C$ of area $OA\bar{V}D$, the value of the firm.

Suppose that the manager of one firm, JV_1, knows that its value exceeds $\bar{V}$ in all states of the world, and is less risky than $A\bar{V}$ as shown by the shallower slope of JV_1. If the manager can communicate this to investors, he will receive a higher price for the shares when they are floated, and his compensation will increase to $\alpha E(V_1)$ shown by M_1EF in Figure 2. Signalling will convey this information to investors but at a cost given by GH. The cost schedule is negatively sloped reflecting the assumption that signalling costs are inversely related to value.

By signalling, the manager's compensation is OM_1FD and he receives GJV_1H from selling shares in the firm. If he does not signal, the manager receives $OA\bar{V}D$ in compensation and share sale proceeds, and he incurs an opportunity loss of $AJV_1\bar{V} - M_1GHF$ which is the manager's net gain from signalling. Hence, he has an incentive to signal.

In Figure 3, $A\bar{V}$ and MBC are again the value and management compensation schedules for the average firm. Suppose that the manager of firm V_2 knows that its value is less than $\bar{V}$ in most states of the world, and that it is a riskier investment as shown by the steeper slope of OV_2. If this information were known to investors, the manager of V_2 would only receive EV_2F when issuing the shares and OEFD in compensation. Uninformed, the market would pay $B\bar{V}C$ for the shares, and the manager's compensation would be OABCD. Hence, the gain to him from investors not being informed is $OAG - GV_2\bar{V}$. No signalling will occur, and uninformed shareholders bear the loss.

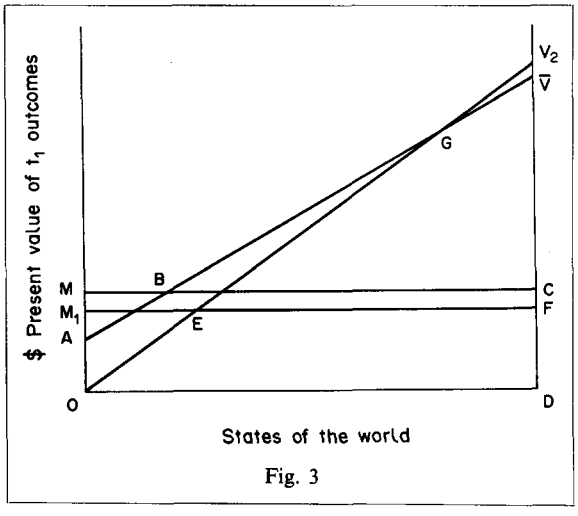


Fig. 3

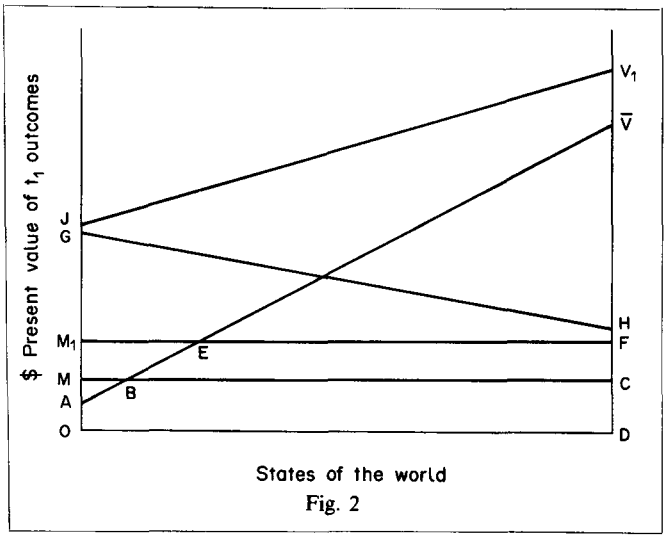


Fig. 2

Similar proofs could be offered for more complex situations, with the same results. The graphical proof also can be adapted easily to show similar effects of information asymmetry where there is debt financing.

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